

45 Days Java Learning Plan

Week 1: Getting Started

Day 1-3: Introduction and Setup

Day 1: Introduction to Java, its history, and uses.

Day 2: Installing JDK, setting up the environment, and choosing an IDE (IntelliJ IDEA/Eclipse).

Day 3: Writing and running your first "Hello, World!" program.

Day 4-7: Basic Syntax

Day 4: Variables and Data Types.

Day 5: Basic Operators (Arithmetic, Relational, Logical).

Day 6: Control Flow Statements (if-else, switch).

Day 7: Loops (for, while, do-while).

Week 2: Object-Oriented Programming (OOP)

Day 8-10: Classes and Objects

Day 8: Introduction to Classes and Objects.

Day 9: Constructors and Methods.

Day 10: The this keyword.

Day 11-14: Advanced OOP Concepts

Day 11: Inheritance.

Day 12: Polymorphism.

Day 13: Abstraction.

Day 14: Encapsulation.

Week 3: Core Java Concepts

Day 15-17: Arrays and Strings

Day 15: Arrays (single-dimensional and multi-dimensional).

Day 16: String class and methods.

Day 17: StringBuilder and StringBuffer.

Day 18-21: Collections Framework

Day 18: Introduction to Collections.

Day 19: List (ArrayList, LinkedList).

Day 20: Set (HashSet, TreeSet).

Day 21: Map (HashMap, TreeMap).

Week 4: Exception Handling and File I/O

Day 22-24: Exception Handling

Day 22: Types of exceptions.

Day 23: try-catch block.

Day 24: throw, throws keywords, and custom exceptions.

Day 25-28: File I/O

Day 25: File class.

Day 26: Reading from and writing to files.

Day 27: BufferedReader and BufferedWriter.

Day 28: Serialization and Deserialization.

Week 5: Advanced Java Concepts

Day 29-31: Multithreading

Day 29: Introduction to Threads.

Day 30: Creating and starting threads (Runnable and Thread class).

Day 31: Thread lifecycle and synchronization.

Day 32-34: Networking

Day 32: Basics of networking.

Day 33: Socket programming (TCP/UDP).

Day 34: URL class.

Week 6: Java Development Tools and Frameworks

Day 35-37: Java Development Tools

Day 35: Maven and Gradle for project management.

Day 36: JUnit for unit testing.

Day 37: Logging frameworks (Log4j, SLF4J).

Day 38-40: Introduction to Java Frameworks

Day 38: Spring Framework basics.

Day 39: Dependency Injection (DI).

Day 40: Spring Boot introduction.

Week 7: Advanced Topics and Project

Day 41-43: Advanced Topics

Day 41: Lambda Expressions and Functional Programming.

Day 42: Stream API.

Day 43: Java 8+ features.

Day 44-45: Mini Project

Day 44: Start a small project to apply learned concepts.

Day 45: Complete and review the project.



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